

Instructor: Date:

Learner: Regional Branch:

Program: Center:

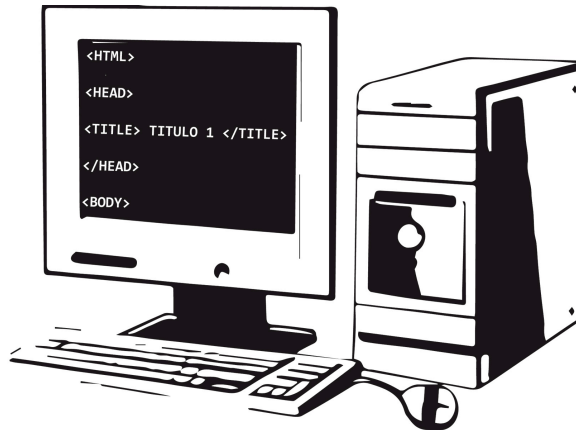
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Let's Start!

The First Programming Languages.



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.
Give information about software history, characteristics and methodologies.

Ask your partner:

- How do you think computers worked before the invention of programming languages?
- How do you imagine computers would work without operating systems?

Can Do Statements:

- I can talk about the first computer languages.
- I can present characteristics for third generation computer languages.
- I can mention information about operating systems in the first electronic computers.



Let's Explore!

Pieces of History of Programming Languages.



Instructions: Read the text below.

Tom: Hey, Carol. What are you doing?

Carol: I am reading about the history of programming languages and operating systems. I have an oral presentation tomorrow. Actually, I think you can help me! Let's pretend you are my audience, OK?

Tom: Sure, go ahead!

Carol: Thanks. Here I go. Did you know that the word software wasn't used before 1958? Actually, at that time, the word computer was not related to electronic machines but for people who work out mathematical problems.

Tom: Wow. I didn't know that.

Carol: In fact, a mathematician called Grace Murray Hopper joined Eckert-Mauchly Computer Corporation to develop an instruction code for their famous computer UNIVAC. She used the term automatic programming to talk about her work.

Tom: Was that code the first programming language?

Carol: No, it wasn't, but her work pointed the way to the development of FORTRAN, which stands for Formula Translator. It was the world's first computer language. Workers from IBM developed it in 1954.

Tom: What did FORTRAN do?

Carol: FORTRAN was a language that made it possible for people to program computers using mathematical formulas and simple English-like instructions. It reduced costs in computer operations. Its development led to another important programming language known as COBOL (Common Business-Oriented Language), and then to a number of other third generation languages, such as ALGOL (Algorithmic Language), BASIC (Beginner's All-purpose Symbolic Instruction Code), or RPG (Report Program Generator).

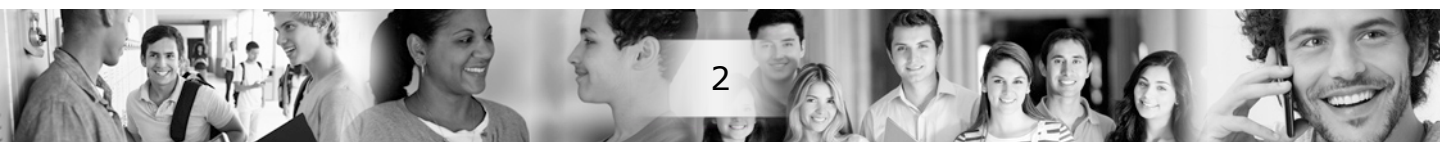
Tom: What are the characteristics of third generation computer languages?

Carol: This kind of languages was not only more efficient but also easier to learn. They became very popular as they standardized program design because of their English-like syntax. 3GLs enabled programs to be hierarchical and structured.

Tom: What about operating systems in the first electronic computers?

Carol: Well, early computers did not have programming languages or operating systems. People had to individually program every function, and there was not any difference between systems and applications software. The first true operating system was MAD, which stands for Michigan Algorithmic Decoder. This operating system was developed at the University of Michigan in the late 50s. It derives from ALGOL; however, the concept of operating systems was not popular until the development of IBM's S/360.

Tom: Wow. That was actually a really fascinating presentation. I'm sure you'll do great!



**Let's Practice!**

Operating Systems in the 1950s.

Instructions: Read the sentences. Underline all the mistakes in the sentences.

1. There aren't any programming languages or operating systems in the first computers.
2. In the early days, there didn't any difference between systems and applications software.
3. The first operating system were MAD.

**Let's Practice!**

The third generation languages are more popular now.

Instructions: Select the correct answer to complete the sentence.

1. In the 40s, the word computer _____ related to electronic machines.
 - a. didn't
 - b. weren't
 - c. wasn't
2. At the beginning, there _____ no differences between systems and applications software.
 - a. was
 - b. wasn't
 - c. were
3. FORTRAN _____ the world's first computer language.
 - a. was
 - b. weren't
 - c. didn't





Extension Activity

Third Generation Languages.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about third generation languages.
- Write three characteristics of a modern third generation language.
- Read that brief text to your teacher. Be careful with the pronunciation.

Self-Assessment

Yes

No

- I look for more information about third generation languages.
- I identify some characteristics of third generation languages.
- I share the information and practice pronunciation in class.

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Target Words

- **derive** (verb): to come from
e.g. This operating system derives from ALGOL.
- **enable** (verb): to give power or authorize someone to do something
e.g. 3GLs enabled programs to be hierarchical and structured.
- **point the way** (expression): to show the way to get somewhere
e.g. Her work pointed the way to the development of FORTRAN.
- **programming** (noun): the process of writing software codes
e.g. I am reading about the history of programming languages and operating systems.
- **true** (adjective): real; genuine or authentic
e.g. The first true operating system was MAD, which stands for Michigan Algorithmic Decoder.



Learning Points

Simple Past Tense of Verb Be - Was in Negative Statements

Use the **simple past** to talk about events that started and finished in the past. You can use the **verb to be** in the **simple past** to describe events in the past about characteristic, places, or locations.

To make a **negative** statement with the **verb to be** in the **simple past**, use the following structure:

Subject + **was / were** + **not** + complement

Example:

COBOL **was not** easy to use.

Use **was** with **I, he, she** and **it**; use **were** with **you, we** and **they**.

You can also use contracted forms:

Examples:

*In the past, computers **weren't** machines but people who worked out mathematical problems.*

*Software **wasn't** a common term in the 50s.*

Examples

- *Did you know that the word software wasn't used before 1958?*
- *The word computer was not related to electronic machines but for people who work out mathematical problems.*
- *There was not any difference between systems and applications software.*

FORTRAN

FORTRAN is a procedural, vital, compiled language with a syntax well suited that represents mathematical formulas.

IBM created **FORTRAN**, which stands for *Formula Translation*, in the 1950s for scientific and engineering applications. This programming language came to dominate the area of programming, and actually, it continues in diverse areas like numerical weather prediction, finite element analysis, computational fluid dynamics, computational physics, crystallography, and computational chemistry.

FORTRAN keeps on incorporating new and better versions. Each has evolved to add extensions to the language and retaining compatibility with prior versions.

